PROGRAM-10

1. Problem Statement

Develop a travel budget analysis system that converts travel expenses from different currencies into a selected base currency using supplied exchange rates. The system should include currency conversion fees, calculate total travel expenses, compare planned spending against the allocated budget, and identify the largest expense category. This helps travelers manage their expenses and avoid exceeding their budgets.

2. Objectives

1. **To accept travel expenses in multiple currencies.**
2. **To convert expenses into a base currency using supplied exchange rates.**
3. **To calculate additional currency conversion fees.**
4. **To calculate the total travel expenditure.**
5. **To compare planned spending with the allocated travel budget.**
6. **To identify the largest expense category.**
7. **To determine whether the traveler is within the budget or has exceeded it.**

3. Key Features

- **Multi-Currency Support: Accept expenses in different currencies, such as USD, EUR, GBP, and INR.**
- **Currency Conversion: Convert expenses into a selected base currency using the given exchange rates.**
- **Conversion Fee Calculation: Include additional fees charged during currency conversion.**
- **Expense Categorization: Organize expenses into accommodation, transportation, food, shopping, and entertainment.**
- **Budget Tracking: Compare total expenses against the planned budget.**
- **Largest Expense Identification: Identify the category that accounts for the highest expenditure.**
- **Remaining Budget Calculation: Display the amount remaining after expenses.**
- **Budget Status Report: Indicate whether the traveler is within budget or over budget.**

4. Applications

1. **Personal travel planning.**

2. **International tourism.**
3. **Business trips and corporate travel.**
4. **Student educational tours.**
5. **Travel agencies and tour operators.**
6. **Family vacation budget management.**
7. **International business expense tracking.**

5. Example Calculation

Suppose a traveler plans a trip with a budget of ₹50,000.

Travel Expenses

Expense Category	Original Amount	Currency	Exchange Rate	Amount in INR
Accommodation	**200**	**USD**	**₹85**	**₹17,000**
Transportation	**100**	**USD**	**₹85**	**₹8,500**
Food	**80**	**EUR**	**₹95**	**₹7,600**
Shopping	**50**	**EUR**	**₹95**	**₹4,750**
Entertainment	**30**	**USD**	**₹85**	**₹2,550**
Total				**₹40,400**

Note: The exchange rates are hypothetical and are provided only for this example.

Conversion Fee

Assume the currency conversion fee is 2% of the converted expense total.

Conversion Fee = ₹40,400 × 2%

Conversion Fee = ₹808

Total Travel Expenditure

Total Expenditure = Converted Expenses + Conversion Fee

Total Expenditure = ₹40,400 + ₹808

Total Expenditure = ₹41,208

Budget Analysis

Description	Amount
Planned Budget	**₹50,000**
Total Expenditure	**₹41,208**

Description	Amount
Remaining Budget	₹8,792
Budget Status	Within Budget

Largest Expense Category

Accommodation costs ₹17,000, making it the largest expense category.

Result: The traveler spends ₹41,208, remains within the ₹50,000 budget, and has ₹8,792 remaining. Accommodation is the largest expense category.

6. Technologies Used

- **Programming Language: Python**
- **User Interface: HTML, CSS, and JavaScript (optional)**
- **Database: MySQL or SQLite (optional)**

7. Expected Outcome

The system will provide an organized analysis of travel expenses in multiple currencies. It will calculate conversion fees, track total spending, identify the largest expense category, and determine whether the travel budget has been exceeded. This will help travelers make informed financial decisions and manage their travel expenses effectively.

10. Campus Event Financial Planning: Calculate venue, equipment, catering, transport, and certificate costs; determine participant contributions, sponsorship shortfalls, and the minimum registration count needed to break even

Here is a simple project explanation for 10. Campus Event Financial Planning, suitable for a college assignment or mini-project.

1. Problem Statement

Develop a campus event financial planning system that calculates the total expenses of organizing a college event, including venue rental, equipment, catering, transportation, and certificates. The system should determine the contribution required from each participant, calculate sponsorship shortfalls, and identify the minimum number of registrations needed to break even. This helps event organizers manage finances effectively and avoid budget deficits.

2. Objectives

1. **To calculate the total cost of organizing a campus event.**

2. **To estimate venue, equipment, catering, transportation, and certificate expenses.**
3. **To calculate the financial contribution required from each participant.**
4. **To determine the sponsorship amount needed to cover event expenses.**
5. **To identify any shortfall between available sponsorship and total costs.**
6. **To calculate the minimum number of registrations required to break even.**
7. **To generate a financial summary to support event planning.**

3. Key Features

- **Venue Cost Calculation: Calculate the cost of renting the event venue.**
- **Equipment Cost Calculation: Include expenses for microphones, projectors, speakers, and other equipment.**
- **Catering Cost Calculation: Calculate food and refreshments based on the number of participants.**
- **Transportation Cost Calculation: Estimate transportation expenses for participants and guests.**
- **Certificate Cost Calculation: Calculate certificate printing costs based on the number of participants.**
- **Participant Contribution: Determine the registration fee required from each participant.**
- **Sponsorship Analysis: Calculate available sponsorship funds and identify funding shortages.**
- **Break-Even Analysis: Determine the minimum number of registrations required to cover event expenses.**
- **Financial Report: Display total costs, sponsorship amounts, registration revenue, and the final financial status.**

4. Applications

1. **College technical symposiums.**
2. **AI and machine learning workshops.**
3. **Cultural festivals and annual college days.**
4. **Student seminars and conferences.**
5. **Sports competitions.**
6. **Hackathons and coding competitions.**
7. **College farewell parties and graduation ceremonies.**

5. Example Calculation

Suppose a college organizes a technical event with the following expenses.

A. Event Expenses

Expense Category	Cost
Venue Rental	₹10,000
Equipment	₹5,000
Catering (100 participants × ₹150)	₹15,000
Transportation	₹3,000
Certificates (100 × ₹20)	₹2,000
Total Event Cost	₹35,000

B. Sponsorship Analysis

Suppose the college receives ₹15,000 in sponsorship.

Sponsorship Shortfall:

Shortfall = Total Event Cost − Available Sponsorship

Shortfall = ₹35,000 − ₹15,000

Sponsorship Shortfall = ₹20,000

The organizers need another ₹20,000 from registration fees or additional sponsors to cover the event costs.

C. Participant Contribution

If 100 participants register:

Registration Fee per Participant = Remaining Cost ÷ Number of Participants

Registration Fee = ₹20,000 ÷ 100

Registration Fee = ₹200 per participant

Therefore, each participant must contribute ₹200 to cover the remaining expenses, assuming the sponsorship is confirmed.

D. Break-Even Analysis

Suppose:

- **Fixed costs (venue, equipment, and transportation) = ₹18,000**
- **Variable cost per participant (catering and certificates) = ₹170**
- **Sponsorship = ₹15,000**

- **Registration fee per participant = ₹200**

Net fixed cost after sponsorship = ₹18,000 − ₹15,000 = ₹3,000.
Each registration contributes ₹200 − ₹170 = ₹30 toward fixed costs.
Result: The event needs at least 100 registrations to break even under these assumptions.

6. Technologies Used

- **Programming Language: Python**
- **User Interface: HTML, CSS, and JavaScript (optional)**
- **Database: MySQL or SQLite (optional)**

7. Expected Outcome

The system will help college event organizers estimate expenses, calculate registration fees, evaluate sponsorship funding, and determine the minimum number of participants required to cover costs. It will support better financial planning, reduce the risk of budget deficits, and help ensure that campus events are financially sustainable.

Programm

```
# Campus Event Financial Planning

# Input event details
event_name = input("Enter event name: ")

venue_cost = float(input("Enter venue cost (Rs.): "))
equipment_cost = float(input("Enter equipment cost (Rs.): "))
catering_per_person = float(input("Enter catering cost per participant (Rs.): "))
transport_cost = float(input("Enter transport cost (Rs.): "))
certificate_per_person = float(input("Enter certificate cost per participant (Rs.): "))

expected_participants = int(input("Enter expected number of participants: "))
registration_fee = float(input("Enter registration fee per participant (Rs.): "))
```

```
sponsorship = float(input("Enter sponsorship amount (Rs.): "))

# Validate inputs
if (venue_cost < 0 or equipment_cost < 0 or
    catering_per_person < 0 or transport_cost < 0 or
    certificate_per_person < 0 or expected_participants <= 0 or
    registration_fee < 0 or sponsorship < 0):

    print("Invalid input. Please enter valid values.")

else:
    # Calculate fixed costs
    fixed_cost = venue_cost + equipment_cost + transport_cost

    # Calculate variable costs
    variable_cost_per_person = catering_per_person +
certificate_per_person
```

```
    total_variable_cost = variable_cost_per_person *
expected_participants

    # Calculate total event cost
    total_cost = fixed_cost + total_variable_cost

    # Calculate participant contributions
    participant_contributions = registration_fee *
expected_participants

    # Calculate total available funds
    total_funds = participant_contributions + sponsorship

    # Calculate sponsorship shortfall
    shortfall = max(0, total_cost - total_funds)

    # Calculate break-even registration count
```

```
    # Each registration contributes the fee minus variable cost
    contribution_per_person = registration_fee -
variable_cost_per_person

    if contribution_per_person > 0:
        required_registrations = max(
            0,
            int((fixed_cost - sponsorship) / contribution_per_person)
            if fixed_cost > sponsorship else 0
        )

        # Round up to the next whole participant if necessary
        if required_registrations * contribution_per_person <
fixed_cost - sponsorship:
            required_registrations += 1

        break_even_possible = True
```

```
    elif sponsorship >= fixed_cost:
        required_registrations = 0
        break_even_possible = True

    else:
        required_registrations = None
        break_even_possible = False

    # Display financial report
    print("\n========== CAMPUS EVENT REPORT
==========")
    print("Event Name:", event_name)
    print("----------------------------------------")
    print("Venue Cost: Rs.", format(venue_cost, ".2f"))
    print("Equipment Cost: Rs.", format(equipment_cost, ".2f"))
    print("Catering Cost: Rs.", format(total_variable_cost -
          certificate_per_person * expected_participants, ".2f"))
```

```
    print("Transport Cost: Rs.", format(transport_cost, ".2f"))
    print("Certificate Cost: Rs.", format(
        certificate_per_person * expected_participants, ".2f"))
    print("----------------------------------------")
    print("Total Event Cost: Rs.", format(total_cost, ".2f"))
    print("Participant Contributions: Rs.",
          format(participant_contributions, ".2f"))
    print("Sponsorship Amount: Rs.", format(sponsorship, ".2f"))
    print("Total Available Funds: Rs.", format(total_funds, ".2f"))
    print("Sponsorship/Funding Shortfall: Rs.", format(shortfall,
".2f"))


    if break_even_possible:
        print("Minimum Registrations to Break Even:",
              required_registrations)
    else:
        print("Break-even is not possible with the given fee and
costs.")
```

```
    if total_funds >= total_cost:
        print("Status: Event is financially feasible.")
    else:
        print("Status: Additional funding is required.")


print("==========================================")
```

output:

```
Enter event name: RRR events
Enter venue cost (Rs.): 50000
Enter equipment cost (Rs.): 50000
Enter catering cost per participant (Rs.): 20000
Enter transport cost (Rs.): 600
Enter certificate cost per participant (Rs.): 621
Enter expected number of participants: 358
Enter registration fee per participant (Rs.): 500
Enter sponsorship amount (Rs.): 5123

========== CAMPUS EVENT REPORT ==========
```

Event Name: RRR events

Venue Cost: Rs. 50000.00

Equipment Cost: Rs. 50000.00

Catering Cost: Rs. 7160000.00

Transport Cost: Rs. 600.00

Certificate Cost: Rs. 222318.00

Total Event Cost: Rs. 7482918.00

Participant Contributions: Rs. 179000.00

Sponsorship Amount: Rs. 5123.00

Total Available Funds: Rs. 184123.00

Sponsorship/Funding Shortfall: Rs. 7298795.00

Break-even is not possible with the given fee and costs.

Status: Additional funding is required.

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